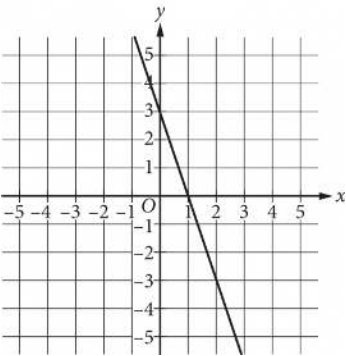


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



What is the equation of the line shown in the xy-plane above?

- Answer
- A. $y = 3x - 3$
- B. $y = -3x + 3$
- C. $y = \frac{1}{3}x - 3$
- D. $y = -\frac{1}{3}x + 3$

Correct Answer: B

Rationale

Choice B is correct. Any line in the xy-plane can be defined by an equation in the form $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. From the graph, the y-intercept of the line is $(0, 3)$. Therefore, $b = 3$. The slope of the line is the change in the value of y divided by the change in the value of x for any two points on the line. The line shown passes through $(0, 3)$ and $(1, 0)$, so $m = \frac{3 - 0}{0 - 1}$, or $m = -3$. Therefore, the equation of the line is $y = -3x + 3$.

Choices A and C are incorrect because the equations given in these choices represent a line with a positive slope. However, the line shown has a negative slope. Choice D is incorrect because the equation given in this choice represents a line with slope of $-\frac{1}{3}$. However, the line shown has a slope of -3 .